# SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

## CSA6303 - Threat Intelligence and Network Security

### A3 - Practical Laboratory Exercises

Laboratory Record

| | |
|---|---|
| **Subject** | Threat Intelligence |
| **Course Outcome** | CO3 |
| **Total Marks** | 30 |
| **Questions** | 3 (all compulsory) |
| **Marks per Question** | 10 |
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| **Faculty** | Dr. V. Parthipan |
| **Slot** | A |

### Instructions

Answer all questions. Support answers with examples and diagrams where appropriate.

CO3: Perform threat analysis using intelligence models and utilize industry tools for monitoring and detection.

### Code of Conduct

I THAMMISETTI RAJU / 192465038 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of Student: _____

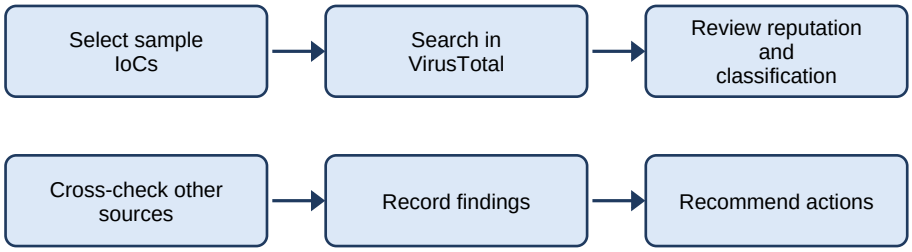
## Exercise 1: Threat Intelligence Platform Investigation [10]

**Objective:** To investigate suspicious IP addresses, domains and file hashes using a threat intelligence platform.

**Tools / platforms:** VirusTotal (primary); cross-checks with AbuseIPDB, AlienVault OTX, URLhaus and MalwareBazaar (all free and authorised for analysis use).

**Note:** the IoCs below are sample values for the lab (IPs are from the reserved documentation ranges 203.0.113.0/24 and 198.51.100.0/24; the domain is a made-up name; the hash is the harmless EICAR antivirus test file). Result values marked *sample* are illustrative. Insert your own screenshots and the exact numbers you see on the platform when you run the lab.

**Procedure**




### Procedure

- Open the platform and use the **Search** box; paste each IP address, domain or file hash (SHA-256) one by one.
- Read the **Detection** tab: number of security vendors flagging the item (for example 12/94), threat label and category (malware, phishing, C2).
- Open **Details** (first/last seen, registrar, ASN, country, file type and names), **Relations** (communicating files, resolutions, URLs) and **Community** comments.
- Cross-check the IP in AbuseIPDB (abuse confidence score) and OTX (pulses), and the file hash in MalwareBazaar.
- Record the results in the table and decide the risk level and action.

*[Screenshot 1: VirusTotal search result / detection tab for the sample IP]*

### Sample IoCs examined and findings

| # | IoC (type) | Reputation / detection (sample) | Classification and context | Risk |
|---|---|---|---|---|
| 1 | 203.0.113.45 (IP) | 14/94 vendors flag as malicious; AbuseIPDB confidence 92% | SSH brute force, scanning; hosted in a VPS network; many abuse reports in 30 days | High |
| 2 | update-check-svc.example (domain) | 9/94 flag as malicious; created 5 days ago | Newly registered domain; C2 / phishing category; resolves to 198.51.100.77 | High |
| 3 | 275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f (SHA-256) | Detected by the large majority of vendors (EICAR test file) | Not real malware: standard antivirus test string; used to verify detection | Informational |
| 4 | 198.51.100.77 (IP) | 5/94 flag; OTX pulse linked to a C2 campaign | Hosts the domain in #2; receives repeated outbound HTTPS from internal hosts | High |

### Analysis

- Indicators 1, 2 and 4 are related: the domain resolves to the IP, and the IP is also reported for scanning, so they probably belong to one attacker infrastructure. Corroboration across independent sources increases confidence.
- A single vendor detection can be a false positive; confidence is high here because several vendors, multiple platforms and community reports agree.
- The EICAR hash is flagged but is harmless, so it is classified as *informational* (shows the detection tools work).

### Recommended security actions

- **Block** IPs 203.0.113.45 and 198.51.100.77 and the domain at the firewall, proxy and DNS (sinkhole).
- **Hunt** in SIEM and firewall logs for any internal host that contacted these IoCs and isolate any such host.
- Enforce account lockout and MFA on SSH/VPN, and review failed-login logs for the attacker IP.
- Add the IoCs to the watch-list / threat-intel feed with an expiry date, and re-check them periodically because indicators age.

### Threat Intelligence Investigation Report (summary)

| Field | Entry |
|---|---|
| Report ID / date / analyst | TI-LAB-01 / 02-Nov-2026 / THAMMISETTI RAJU (192465038) |
| Scope and sources | 4 IoCs; VirusTotal, AbuseIPDB, OTX, MalwareBazaar |
| Key findings | 3 high-risk related indicators (IP, domain, IP), 1 benign test hash |
| Confidence | High for IoCs 1, 2, 4 (multiple sources agree) |
| Actions | Block, hunt, harden authentication, monitor |

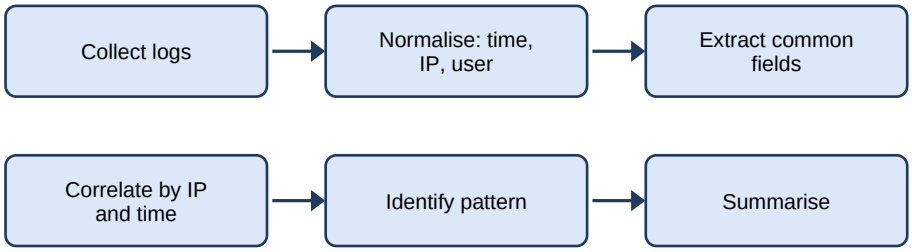
**Result:** the sample IoCs were investigated with a threat-intelligence platform, risk-rated, and appropriate security actions were recommended.

## Exercise 2: Security Log Collection and Correlation [10]

**Objective:** To collect and correlate security events from different sources.

**Tools:** sample authentication log (Linux *auth.log*), firewall log and web server (Apache) access log; Linux commands (grep, awk, sort, uniq) or a SIEM such as Splunk / ELK. All IPs are sample documentation addresses.

**Procedure**




### Step 1: Sample logs collected

**(a) Authentication log** - /var/log/auth.log

```
Nov  2 02:14:03 srv1 sshd[1201]: Failed password for admin from 203.0.113.45 port 51022
Nov  2 02:14:06 srv1 sshd[1201]: Failed password for admin from 203.0.113.45 port 51023
Nov  2 02:14:09 srv1 sshd[1201]: Failed password for admin from 203.0.113.45 port 51024
Nov  2 02:14:12 srv1 sshd[1201]: Failed password for admin from 203.0.113.45 port 51025
Nov  2 02:14:15 srv1 sshd[1201]: Failed password for admin from 203.0.113.45 port 51026
Nov  2 02:15:02 srv1 sshd[1209]: Accepted password for admin from 203.0.113.45 port 51030
Nov  2 09:31:40 srv1 sshd[1388]: Accepted password for priya from 10.0.0.31 port 40022
```

**(b) Firewall log**

```
02:13:50 DENY  TCP 203.0.113.45:51001 -> 10.0.0.5:23
02:13:52 DENY  TCP 203.0.113.45:51002 -> 10.0.0.5:445
02:13:54 DENY  TCP 203.0.113.45:51003 -> 10.0.0.5:3389
02:14:01 ALLOW TCP 203.0.113.45:51022 -> 10.0.0.5:22
02:15:40 ALLOW TCP 203.0.113.45:51044 -> 10.0.0.5:80
02:17:30 ALLOW TCP 10.0.0.5:49832 -> 198.51.100.77:443   bytes_out=2.1MB
```

**(c) Web server access log**

```
203.0.113.45 [02:15:41] "GET /admin/login.php HTTP/1.1" 200 1532
203.0.113.45 [02:16:20] "POST /upload.php HTTP/1.1" 200 88
203.0.113.45 [02:16:55] "GET /uploads/shell.php?cmd=whoami HTTP/1.1" 200 12
10.0.0.31    [09:35:12] "GET /index.html HTTP/1.1" 200 4120
```

*[Screenshot 2: log files opened in terminal / Splunk / Kibana]*

### Step 2: Identify common IPs, timestamps and user activities

```
grep 'Failed password' auth.log | awk '{print $11}' | sort | uniq -c | sort -nr
#   5 203.0.113.45

cat auth.log fw.log access.log | grep -o '203.0.113.45' | wc -l
```

| Item | Observation |
|---|---|
| Common IP | 203.0.113.45 appears in all three sources (authentication, firewall, web server) |
| Timestamps | Activity is concentrated between 02:13 and 02:17 (outside working hours) |

| Item | Observation |
|---|---|
| User activity | Account *admin*: 5 failed logins then a successful login from the same external IP; normal user *priya* logs in from internal 10.0.0.31 at 09:31 (benign) |
| Target | 10.0.0.5 (server srv1): ports 23, 445, 3389 denied, 22 and 80 allowed |

### Step 3: Correlated event timeline

| Time | Source | Event | Meaning / ATT&CK |
|---|---|---|---|
| 02:13:50-54 | Firewall | Denied connections to ports 23, 445, 3389 | Port scan / reconnaissance (T1046) |
| 02:14:03-15 | Auth log | 5 failed SSH logins for admin | Brute force (T1110) |
| 02:15:02 | Auth log | Successful admin login from the same IP | Valid accounts (T1078): compromise |
| 02:16:20 | Web log | POST /upload.php | File upload |
| 02:16:55 | Web log | shell.php?cmd=whoami | Web shell execution (T1505.003) |
| 02:17:30 | Firewall | srv1 sends 2.1 MB to 198.51.100.77:443 | C2 / exfiltration (T1071, T1041) |

### Correlated Security Event Summary

Individually, the failed logins, denied connections and a web upload may look routine. Correlated by **source IP and time**, they show one attack chain: **scan -> brute force -> successful login -> web shell -> outbound C2 traffic**. Server *srv1 (10.0.0.5)* should be treated as **compromised (severity: High)**.

### Recommended actions

- Block 203.0.113.45 and 198.51.100.77; isolate srv1; remove shell.php and the upload vulnerability.
- Reset the admin password, disable direct admin SSH from the internet, enforce key-based login/MFA and fail2ban.
- Create a SIEM correlation rule: repeated failed logins + success + web upload from the same IP within 10 minutes.

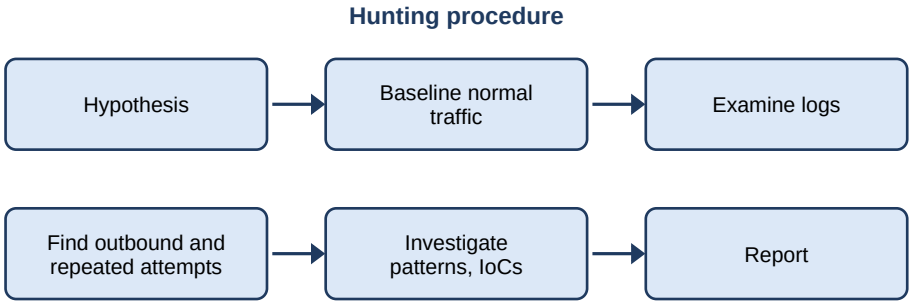
**Result:** logs from three sources were collected, common fields extracted and correlated, and a suspicious attack pattern was identified.

## Exercise 3: Threat Hunting Using Network Logs [10]

**Objective:** To identify suspicious network activities through proactive threat hunting.

**Tools:** sample network flow/connection logs (Zeek *conn.log* / NetFlow style), Wireshark, Linux command line, optional Splunk or ELK. Addresses are sample documentation IPs.

**Hypothesis:** *an internal host is infected and communicates with an external command-and-control server, and an attacker is repeatedly probing our services.*




### Sample network log (simplified)

| Time | Src IP | Dst IP:Port | Proto | Bytes out | Notes |
|---|---|---|---|---|---|
| 09:00:00 | 10.0.0.23 | 198.51.100.77:443 | TCP | 612 | Connection #1 |
| 09:01:00 | 10.0.0.23 | 198.51.100.77:443 | TCP | 612 | Connection #2 (60 s later) |
| 09:02:00 | 10.0.0.23 | 198.51.100.77:443 | TCP | 612 | Connection #3 (60 s later) |
| ... every 60 s | 10.0.0.23 | 198.51.100.77:443 | TCP | 612 | 120 connections in 2 hours |
| 02:20:00 | 10.0.0.5 | 198.51.100.77:443 | TCP | 480 MB | Large night upload |
| 03:01:10-50 | 203.0.113.45 | 10.0.0.5:22 | TCP | SYN only | 250 attempts in 1 minute |
| 10:12:05 | 10.0.0.23 | a8f3k2x9q1z7w5.update-check-svc.example:53 | UDP | 180 | Very long DNS name |

### Hunting steps and commands

```
# 1. Top external destinations (stacking)
awk '{print $5}' conn.log | sort | uniq -c | sort -nr | head

# 2. Beaconing: time gaps for one host/destination pair
grep '10.0.0.23.*198.51.100.77' conn.log | awk '{print $1}' | sort | head -20

# 3. Repeated connection attempts (SYN scan / brute force)
awk '{print $3}' conn.log | sort | uniq -c | sort -nr | head

# Wireshark display filters
ip.addr==198.51.100.77 && tcp.port==443
tcp.flags.syn==1 && tcp.flags.ack==0
len(dns.qry.name) > 40
```

*[Screenshot 3: Wireshark / log search showing periodic connections to 198.51.100.77]*

### Findings

| # | Finding | Evidence | Assessment |
|---|---|---|---|
| 1 | Beaconing from 10.0.0.23 to 198.51.100.77:443 | 120 connections at fixed 60 s intervals; identical 612-byte packets | Likely malware C2 channel (T1071.001) |
| 2 | Unusual large outbound transfer | 10.0.0.5 sent 480 MB to the same external IP at 02:20 | Possible data exfiltration (T1041); server normally sends little data |
| 3 | Repeated connection attempts | 203.0.113.45 made 250 SYN attempts to port 22 within a minute | Brute force / scanning (T1110, T1046) |
| 4 | Suspicious DNS | Long random sub-domain of update-check-svc.example | DGA or DNS tunnelling (T1568 / T1572) |

### Indicators of Compromise (IoCs) documented

| Type | Value | Context |
|---|---|---|
| IP address | 198.51.100.77 | Suspected C2 and exfiltration destination |
| IP address | 203.0.113.45 | Source of brute-force / scanning |
| Domain | update-check-svc.example | Contacted by 10.0.0.23, long random sub-domains |
| Pattern | 60-second periodic HTTPS connections, 612 bytes | Beacon signature for detection rule |
| Affected hosts | 10.0.0.23 (workstation), 10.0.0.5 (server) | Require investigation and isolation |

### Recommendations

- **Contain:** isolate 10.0.0.23 and 10.0.0.5; block the IPs and domain on the firewall, proxy and DNS.
- **Investigate:** collect EDR/memory data from both hosts, look for the malicious process, persistence and dropped files, and hunt the IoCs on all other hosts.
- **Harden:** egress filtering so servers cannot connect to the internet freely; restrict SSH to VPN/management network; rate limiting and fail2ban; MFA.
- **Detect:** SIEM alerts for periodic beaconing, large after-hours uploads, high SYN rate and long DNS names; DLP and regular threat hunts.
- Document in the incident ticket and share the IoCs with the SOC.

**Result:** suspicious outbound connections, repeated connection attempts and abnormal DNS were identified in the network logs through threat hunting, IoCs were documented and recommendations were made.